SAFE TEST PLAN · MONITOR V2.0

Bringing the D1 monitor into service, one safe step at a time

The monitor only listens: it has no way to send the D1 anything, so **no step here can change what the D1 does**. The risks are the usual ones of working near batteries, and a wiring mistake. Each stage checks one thing before the next. Stop at the first thing that looks wrong.

At every step the Cerbo sends nothing: it can't change what the D1 does



What protects you

- The D1 protects itself (D1 manual). It stops if the 12 V side goes over 16–18 V or under 8 V, the 48 V side goes over 58–60 V or under 24 V, the current goes over 213 A, there's a short, or it gets hotter than 248 °F (120 °C) inside
- Your BMS and fuses on HV+, LV+ and KL30
- The cable only carries data, and the monitor never transmits on it

Before you start

- Fuses on HV+ and LV+ (D1 manual), and 3 A on KL30
- A multimeter, plus a DC clamp meter if you have one
- The D1 switched off (KL30/KL15) while you wire
- The Installation Guide (`docs/installation-guide.pdf`) at hand

Stage	What	Risk
1	Simulation on the Cerbo	None: no converter, no CAN bus
2	Wiring check, everything off	None: no power
3	Listen	None: the Cerbo sends nothing
4	Installed and watching for a day	None: the Cerbo sends nothing

1 Simulation: no converter at all

In Cerbo Setup, choose **9**. The real monitor runs on your Cerbo against a pretend D1, and no CAN socket is ever opened. A real installation keeps running beside it.

Check	OK
"DC-DC D1 (SIMULATED)" cards appear in the Cerbo's switch pane, and in VRM's Switches	
Its web page opens at **http://<Cerbo IP>:8089**, and the charts start filling after a minute	
The pretend 12 V side drains to 12.5 V, then Status shows **Autonomous** and it charges back to about 13.4 V (12 V side current +, 48 V side −)	
Every three minutes a pretend alternator pushes the 12 V side above 13.5 V and the pretend D1 boosts (12 V side current −, 48 V side +) until 12.3 V	

Stop with Ctrl+C. The simulated cards disappear. The pretend D1 follows the manual's logic, but it can't tell you anything about the real one.

2 Wiring check, everything off

Check	OK
Continuity: RJ45 pin 7 to white/brown, pin 8 to brown (guide step 2b)	
No continuity (no beep) between white/brown or brown and LV+, HV+ or GND	
Lever connector A holds the D1's CAN_H wire, white/brown and one resistor leg; B holds CAN_L, brown and the other leg. Every wire fully in, levers closed, resistor legs insulated. White/green and the other unused wires insulated (no ground wire)	
RJ45 not in the Cerbo: white/brown to brown reads about 120 Ω (your resistor at the D1)	
RJ45 in the Cerbo, with the VE.Can terminator in the other socket: about 60 Ω	

3 Listen

Power the D1 as normal (KL30 and KL15 on). In Cerbo Setup choose **2** (find), then **3** (watch).

Check	What you saw	OK
Converter found, at the right speed (if it reports bus errors, switch speed as the guide says)		
12 V side reading within about 0.2 V of your meter on the D1's LV+ to GND		
48 V side reading within about 0.5 V of your meter on HV+ to GND		
While the D1 is converting: which side shows + current? Check with the clamp meter which way the current really flows		
No alarms, or only alarms that make sense (for example low voltage when the battery really is low)		

4 Installed and watching

Choose **4** (install). Leave it running for a day.

Check	OK
Three cards on the Cerbo, in VRM and on the web page, with sensible readings	
The web page's charts cover the day, with no gaps you can't explain (a gap means the D1 went quiet, or the Cerbo restarted)	
The D1 behaves exactly as before: buck when the 12 V side is low, boost when it's high, standby otherwise	
After a Cerbo reboot the monitor comes back by itself (option 5 shows "Converter online")	

> **Please share what you found:** which way the current sign went in stage 3, and anything that didn't match. Open an issue on the project's GitHub page.

Tested and signed off by: _____ Date: _____